Pratyush Kiran Rath

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EDUCATION

Centurion University of Technology and Management (CUTM) Odisha, India

Bachelor of Technology in Computer Science and Engineering 2025 – 2029

Relevant Coursework: Systems Programming, Data Structures & Algorithms, Operating Systems, Computer Organization, Database Systems.

FEATURED PROJECTS & RESEARCH

Transformer Inference Profiler & Memory Telemetry Suite Python, PyTorch, Plotly, tmpfs, PyTest

Open Source Mechanistic Telemetry Suite github.com/Pratyush-kya/transformer-inference-profiler

- Architected an in-RAM zero-disk telemetry engine utilizing Linux `/dev/shm` tmpfs ring buffers and in-memory SQLite, achieving a verified **0.00 MB/s disk write rate** during continuous LLM decoding.
- Implemented analytical KV-cache byte accounting for Multi-Head (MHA), Grouped-Query (GQA), and Multi-Head Latent Attention (MLA).
- Intercepted logits to compute instantaneous Shannon entropy $H(X)$ to detect token candidate uncertainty in real time.
- Validated system reliability with a **29/29 PyTest test suite** and streaming 60 FPS Plotly Dash visual dashboard.

Resumint — Privacy-First Client-Side ATS Resume Analyzer Next.js 16, React 19, TypeScript, pdfjs-dist

Production Client-Side Web Application resumint-eta.vercel.app

- Built a zero-backend, client-side resume scoring application using Next.js static edge deployment and Web Workers.
- Integrated in-browser PDF text extraction via `pdfjs-dist`, ensuring **0 bytes of user data** are ever uploaded to remote servers.
- Delivered sub-250ms keyword extraction and heuristic semantic matching with zero server infrastructure overhead.

Arogya-Relay — Emergency Ambulance Dispatch & Telemetry Next.js, Supabase PostgreSQL 17, TypeScript

Healthcare Prototype & State Machine arogya-relay.vercel.app

- Designed an atomic dispatch state machine on Supabase PostgreSQL 17 with Row Level Security

(RLS) constraints.

- Achieved sub-150ms state synchronization latency under simulated unstable 3G network conditions.

Operational Workflow Automation Engines n8n, Docker, Webhooks, Bash

Self-Hosted Operational Pipelines github.com/Pratyush-kya/enterprise-n8n-automations

- Constructed webhook-triggered pipelines for automated developer inquiry routing and multi-channel technical note formatting.
- Implemented automated service health monitors with instant push notifications to Telegram.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Bash/Shell, C/C++ (Foundational).

Systems & AI: PyTorch (Logits Interception, KV-Cache Profiling), Linux `/dev/shm`, In-Memory SQLite, PyTest, Plotly Dash.

Frontend & Web: Next.js 16 (App Router, SSG), React 19, Tailwind CSS v4, Web Workers (pdfjs-dist), HTML5/CSS3.

Backend & Tools: Supabase PostgreSQL 17, Docker, n8n Workflow Automation, Git, Linux/Unix CLI, Vercel Edge.